BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project Report

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Tools & Technologies:

- Python
- Pandas
- NumPy
- SQLite
- Power BI
- Git & GitHub

Project Duration:

May 2026 – June 2026

Submission Date:

12 June 2026

Executive Summary

The Bluestock Mutual Fund Analytics project is a comprehensive data analytics solution developed to analyze mutual fund performance, investor behavior, portfolio allocation, and investment risk. The project integrates data engineering, exploratory data analysis, statistical modeling, and business intelligence techniques to generate actionable insights for investors and fund managers.

The primary objective of the project is to transform raw mutual fund data into meaningful information through a structured analytics pipeline. Multiple datasets related to fund performance, net asset values, assets under management, investor transactions, SIP inflows, portfolio holdings, and benchmark indices were collected and processed.

Python was used extensively for data cleaning, transformation, feature engineering, and advanced analytics. SQLite served as the centralized database for storing and managing cleaned datasets. Power BI dashboards were developed to provide interactive visualizations and performance monitoring capabilities.

Advanced analytical techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation Engine, and Portfolio Concentration Analysis using the Herfindahl-Hirschman Index (HHI) were implemented to evaluate risk and investment quality.

The project demonstrates a complete end-to-end analytics workflow and provides valuable insights that can assist investors in making informed financial decisions.

Introduction and Problem Statement

Mutual funds have become one of the most popular investment instruments due to their diversification benefits and professional fund management. However, investors often face challenges in evaluating fund performance, understanding risk levels, and selecting suitable investment options based on their risk appetite.

The availability of large volumes of mutual fund data creates an opportunity to apply data analytics techniques for better decision-making. By analyzing historical fund performance, investor transactions, and portfolio composition, it is possible to identify trends, measure risk, and generate actionable recommendations.

Problem Statement

Investors require a comprehensive platform that can:

- Analyze mutual fund performance effectively.
- Evaluate risk-adjusted returns.
- Understand investor investment behavior.
- Monitor SIP continuity.
- Measure portfolio concentration risk.
- Generate suitable fund recommendations.

Traditional reports often provide fragmented information and lack integrated analytics capabilities. This project addresses this limitation by building a unified mutual fund analytics platform.

Project Objectives

1. To collect and integrate mutual fund datasets from multiple sources.
2. To design and implement a robust ETL pipeline for data preparation.
3. To store and manage processed datasets using SQLite.
4. To perform exploratory data analysis on fund performance and investor behavior.
5. To calculate key performance metrics including returns, volatility, Sharpe Ratio, Alpha, and Beta.
6. To develop advanced risk metrics such as VaR and CVaR.
7. To analyze investor cohorts and SIP continuity behavior.
8. To measure portfolio concentration using the Herfindahl-Hirschman Index.
9. To develop a simple fund recommendation engine.
10. To create interactive dashboards using Power BI.
11. To generate business insights and recommendations.

Expected Outcomes

- Improved understanding of mutual fund performance.
- Better assessment of investment risk.
- Enhanced investor behavior analysis.
- Data-driven investment recommendations.
- Interactive business intelligence dashboards.

Data Sources

Data Sources:

The project uses multiple datasets representing various aspects of mutual fund operations and investor activity.

ETL Design

ETL Workflow Diagram

RAW DATA SOURCES
(Fund Master, NAV History, AUM Data, SIP Inflows, Category Inflows, Investor Transactions, Portfolio Holdings, Benchmark Indices)

↓

DATA EXTRACTION
(Python + Pandas using read_csv())

↓

DATA CLEANING
• Missing Value Handling
• Duplicate Removal
• Data Type Conversion
• Column Standardization
• Data Validation

↓

DATA TRANSFORMATION
• Return Calculations
• Risk Metrics Preparation

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QUALITY CHECKS
• Null Value Checks
• Consistency Validation
• Integrity Verification

↓

SQLITE DATABASE
(bluestock_mf.db)

↓

ANALYTICS LAYER
• Performance Analytics
• Risk Analytics
• Investor Cohort Analysis
• SIP Continuity Analysis
• HHI Concentration Analysis
• Fund Recommendation System

↓

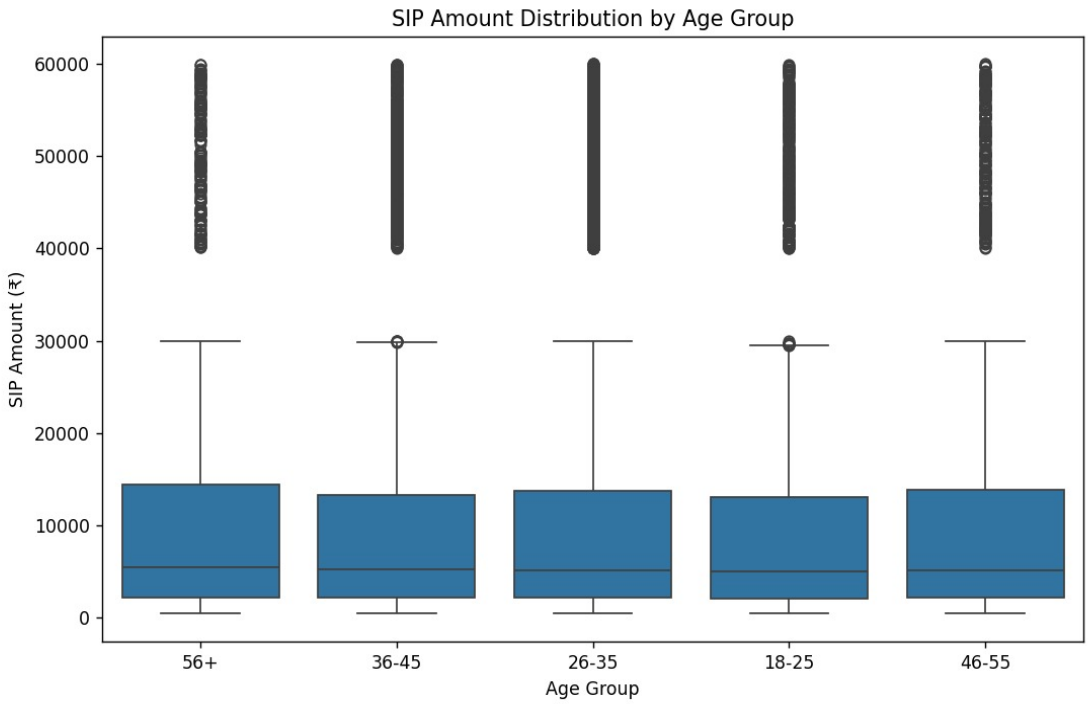
VISUALIZATION LAYER
• Power BI Dashboard
• Analytical Reports
• Business Insights

Exploratory Data Analysis (EDA)
Fund Performance Analysis

The NAV trend analysis was conducted using historical NAV records across multiple mutual fund schemes. The analysis indicates that equity-oriented funds experienced consistent long-term growth despite short-term market fluctuations.

Small-cap and mid-cap funds demonstrated higher volatility compared to large-cap funds. However, they also showed greater return potential over longer investment horizons.

Investor Analysis (EDA)



The investor transaction dataset was analyzed to understand demographic patterns and investment behavior. Variables such as age group, gender, city tier, and annual income were evaluated to identify trends in mutual fund participation.

The analysis indicates that investment activity is concentrated among working-age investors, who contribute the majority of SIP and lump-sum investments. Urban investors from Tier-1 cities showed higher participation levels compared to smaller cities, reflecting greater financial awareness and accessibility to investment products.

Demographic analysis helps fund managers understand investor preferences and design products that better align with customer needs.

Performance Analytics

Performance Metrics of Mutual Fund Schemes:

	amfi_code	cagr_1y	cagr_3y	cagr_5y	cagr_rank	sharpe_ratio	sortino_ratio	alpha	beta	alpha_rank	max_drawdown	drawdown_rank	tracking_error
0	100016	-0.022243	0.012926	0.023168	35.0	-0.201517	-0.351047	0.037476	-0.058268	39.0	-0.247344	34.0	0.199284
1	100025	0.037050	0.039164	0.039127	34.0	-0.567095	-0.941821	0.042818	0.001158	38.0	-0.043083	4.0	0.134535
2	100033	0.532324	0.324425	0.260741	4.0	1.093699	1.829134	0.271954	0.005104	6.0	-0.162172	20.0	0.228699
3	101206	0.479241	0.289677	0.204427	9.0	1.027213	1.799563	0.213998	0.021086	12.0	-0.112916	9.0	0.192706
4	101207	-0.239860	-0.041524	0.069533	39.0	0.162661	0.276644	0.108971	-0.065289	27.0	-0.354469	38.0	0.292117

Performance analytics was conducted to evaluate the risk-adjusted returns and overall effectiveness of mutual fund schemes. Several financial metrics including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error were analyzed to assess fund performance.

The CAGR metrics indicate the annualized growth achieved by funds over different investment horizons. Higher CAGR values generally reflect stronger long-term performance. Risk-adjusted metrics such as Sharpe Ratio and Sortino Ratio were used to determine how efficiently funds generated returns relative to the risks undertaken.

a and Beta were calculated to evaluate fund manager performance and market sensitivity. Alpha measures excess returns generated beyond benchmark expectations, while Beta indicates a fund's responsiveness to market.

Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Historical VaR (95%) and CVaR (95%) Analysis:

	scheme_name	VaR95	CVaR95
0	HDFC Top 100 Fund - Regular Plan - Growth	-0.014364	-0.018060
1	HDFC Short Term Debt Fund - Regular - Growth	-0.003793	-0.004994
2	HDFC Mid-Cap Opportunities Fund - Regular - Gr...	-0.019034	-0.023456
3	ABSL Frontline Equity Fund - Regular - Growth	-0.013282	-0.017439
4	ABSL Small Cap Fund - Regular - Growth	-0.026021	-0.032459

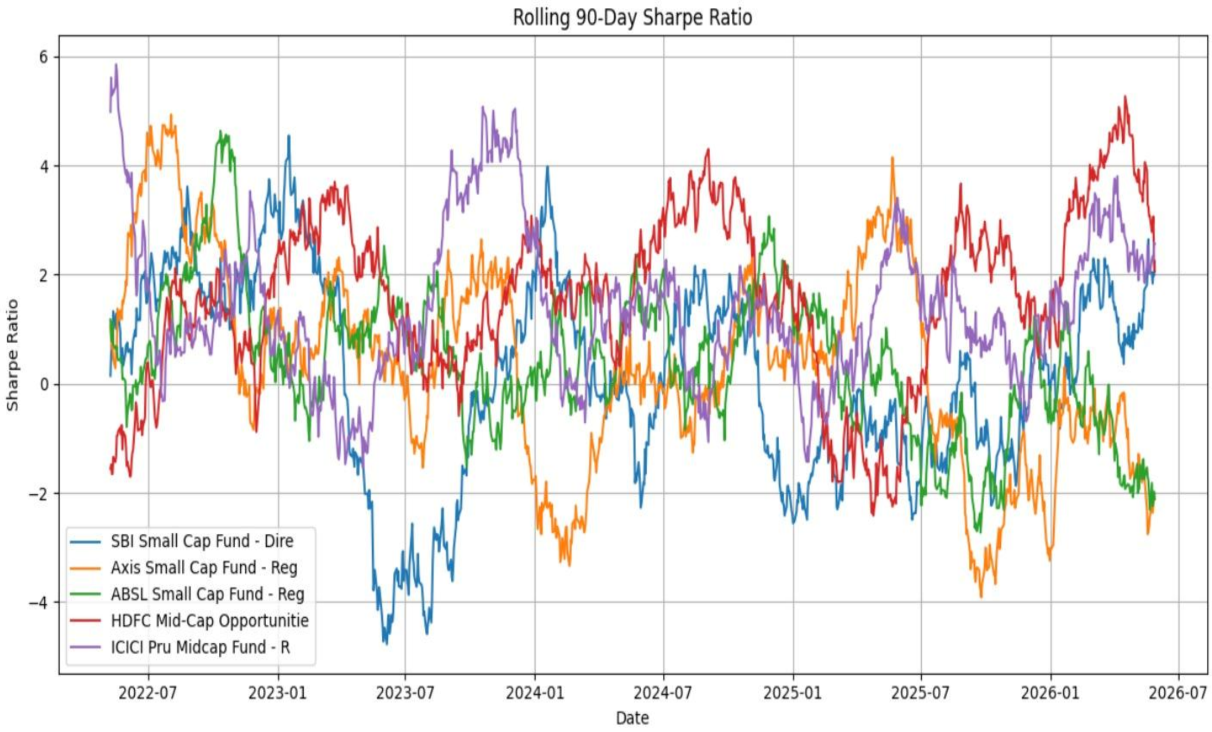
To evaluate downside risk exposure, Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated using daily mutual fund return distributions. These metrics help quantify the potential losses that investors may face under adverse market conditions.

The 95% Historical VaR estimates the maximum expected loss on a typical trading day with 95% confidence. CVaR extends this analysis by calculating the average loss beyond the VaR threshold, providing a more comprehensive measure of extreme downside risk.

The analysis indicates that small-cap and mid-cap funds generally exhibit higher risk levels than large-cap and debt-oriented funds. Funds with higher growth potential often experience greater volatility and larger downside movements during unfavorable market conditions.

Rolling 90-Day Sharpe Ratio Analysis

Rolling 90-Day Sharpe Ratio Analysis for Selected Mutual Fund Schemes



The Rolling 90-Day Sharpe Ratio was calculated to evaluate the consistency of risk-adjusted returns over time. Unlike a single Sharpe Ratio value, the rolling approach measures performance across consecutive 90-day windows, providing a dynamic view of fund behavior under changing market conditions.

The Sharpe Ratio compares excess returns generated by a fund relative to the volatility experienced during the same period. Higher Sharpe Ratio values indicate better risk-adjusted performance, while lower or negative values suggest that the fund is not adequately compensating investors for the risks undertaken.

The rolling analysis revealed noticeable fluctuations across the selected mutual fund schemes. Market volatility, economic events, and sector-specific movements contributed to changes in risk-adjusted performance over time. Funds that maintained consistently positive Sharpe Ratios demonstrated stronger resilience and superior performance stability.

Investor Cohort Analysis

Investor Cohort Analysis Based on Investment Year:

	investor_id	transaction_date	amfi_code	transaction_type	amount_inr	state	city	city_tier	age_group	gender	annual_income_lakh	payment_mode	kyc_status	cohort_year
0	INV003054	2024-01-01	119092	Sip	1834	Telangana	Hyderabad	T30	56+	Female	77.1	UPI	Verified	2024
1	INV002952	2024-01-01	148567	Redemption	392882	Punjab	Amritsar	B30	18-25	Male	7.1	Cheque	Verified	2024
2	INV003420	2024-01-01	118636	Sip	912	Haryana	Faridabad	B30	36-45	Male	47.2	Mandate	Verified	2024
3	INV003436	2024-01-01	118634	Sip	1102	Maharashtra	Mumbai	T30	36-45	Female	54.4	Cheque	Pending	2024
4	INV004691	2024-01-01	119094	Lumpsum	8682	Delhi	Noida	T30	26-35	Male	14.5	Net Banking	Pending	2024

Investor cohort analysis was performed to understand investment behavior across different groups of investors based on their first transaction year. Cohorts enable the identification of long-term investment trends, participation levels, and differences in investment activity between investor groups.

The analysis revealed that investors entering the market in different years exhibited varying investment patterns. By grouping investors according to their entry year, it becomes possible to evaluate average investment amounts, total capital contributions, and overall participation trends.

The cohort analysis indicated that the 2024 investor cohort contributed the highest total investment volume, reflecting stronger participation and investment activity during that period. The average investment amount remained relatively consistent across cohorts, suggesting similar investment behavior among participating investors.

SIP Continuity Analysis

```python
sip['gap_days'] = sip.groupby(
    'investor_id'
)['transaction_date'].diff().dt.days

sip.head()
```

investor_id	transaction_date	amfi_code	transaction_type	amount_inr	state	city	city_tier	age_group	gender	annual_income_lakh	payment_mode	kyc_status	cohort_year	gap_days

Investors with 6+ SIP Transactions

Systematic Investment Plans (SIPs) are one of the most popular investment methods in mutual funds due to their disciplined and long-term investment approach. To evaluate investor consistency, a SIP continuity analysis was performed on investors having six or more SIP transactions.

The average gap between consecutive SIP transactions was calculated for each investor. Investors with an average transaction gap greater than 35 days were classified as at-risk investors, indicating potential discontinuation of their SIP investments.

The analysis helps identify investors who may require additional engagement or support to maintain consistent investment behavior. SIP continuity is an important indicator of investor commitment and long-term wealth creation potential.

Portfolio Concentration Analysis using HHI

scheme_name	category	HHI
Axis Bluechip Fund - Regular - Growth	Large Cap	0.206448
ABSL Small Cap Fund - Regular - Growth	Small Cap	0.200700
SBI Small Cap Fund - Direct Plan - Growth	Small Cap	0.174751
UTI Nifty 50 Index Fund - Regular - Growth	Index	0.174709
Nippon India Large Cap Fund - Regular - Growth	Large Cap	0.168298
Mirae Asset Emerging Bluechip Fund - Regular -...	Large & Mid Cap	0.167930
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	0.157570
ICICI Pru Value Discovery Fund - Regular - Growth	Value	0.153794
HDFC Mid-Cap Opportunities Fund - Direct - Growth	Mid Cap	0.152414
Kotak Bluechip Fund - Regular - Growth	Large Cap	0.149680

Portfolio diversification plays a critical role in risk management and long-term investment performance. To evaluate portfolio concentration levels, the Herfindahl-Hirschman Index (HHI) was calculated for all mutual fund schemes using portfolio holding weights.

The HHI is a widely used concentration metric computed as the sum of squared portfolio weights. Higher HHI values indicate greater concentration in a smaller number of holdings, while lower values indicate better diversification across multiple assets.

The analysis revealed significant differences in concentration levels across fund categories. Certain large-cap and small-cap funds demonstrated relatively high HHI values, indicating greater concentration risk. More diversified funds exhibited lower HHI values and reduced dependency on a limited number of portfolio holdings.

Dashboard Overview and KPI Monitoring

An interactive Power BI dashboard was developed to provide a comprehensive overview of mutual fund performance and investment activity. The dashboard consolidates key metrics into a single interface, enabling users to monitor performance indicators and make data-driven decisions.

The overview dashboard displays critical KPIs such as Total Assets Under Management (AUM), Total Number of Schemes, Average Returns, Investor Participation, and Risk Metrics. Interactive filters allow users to analyze specific fund categories, fund houses, and investment periods.

The dashboard enhances data accessibility by transforming complex analytical results into intuitive visualizations. Decision-makers can quickly identify trends, monitor fund performance, and evaluate investment opportunities through a user-friendly interface.

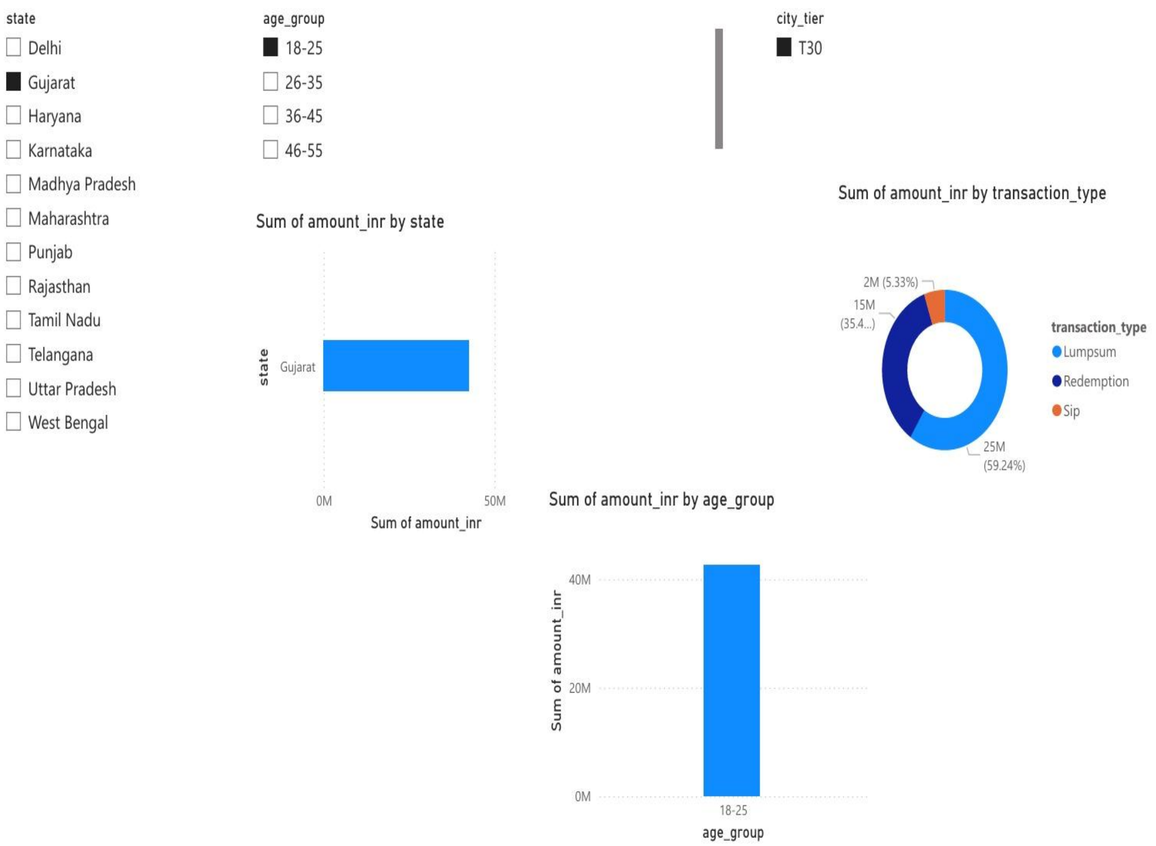
Fund Performance Dashboard

The Fund Performance Dashboard was developed to provide detailed insights into the historical performance of mutual fund schemes across different categories. The dashboard enables users to compare funds using key performance indicators such as returns, Sharpe Ratio, Alpha, Beta, Standard Deviation, and Maximum Drawdown.

Interactive visualizations allow investors to analyze fund performance over multiple investment horizons, including 1-year, 3-year, and 5-year periods. The dashboard supports comparative analysis across fund houses and categories, helping users identify consistently performing schemes.

Performance metrics are presented through charts, tables, and KPI cards, allowing users to evaluate both return potential and associated risk. This comprehensive view supports informed investment decision-making and portfolio construction.

Investor Analysis Dashboard

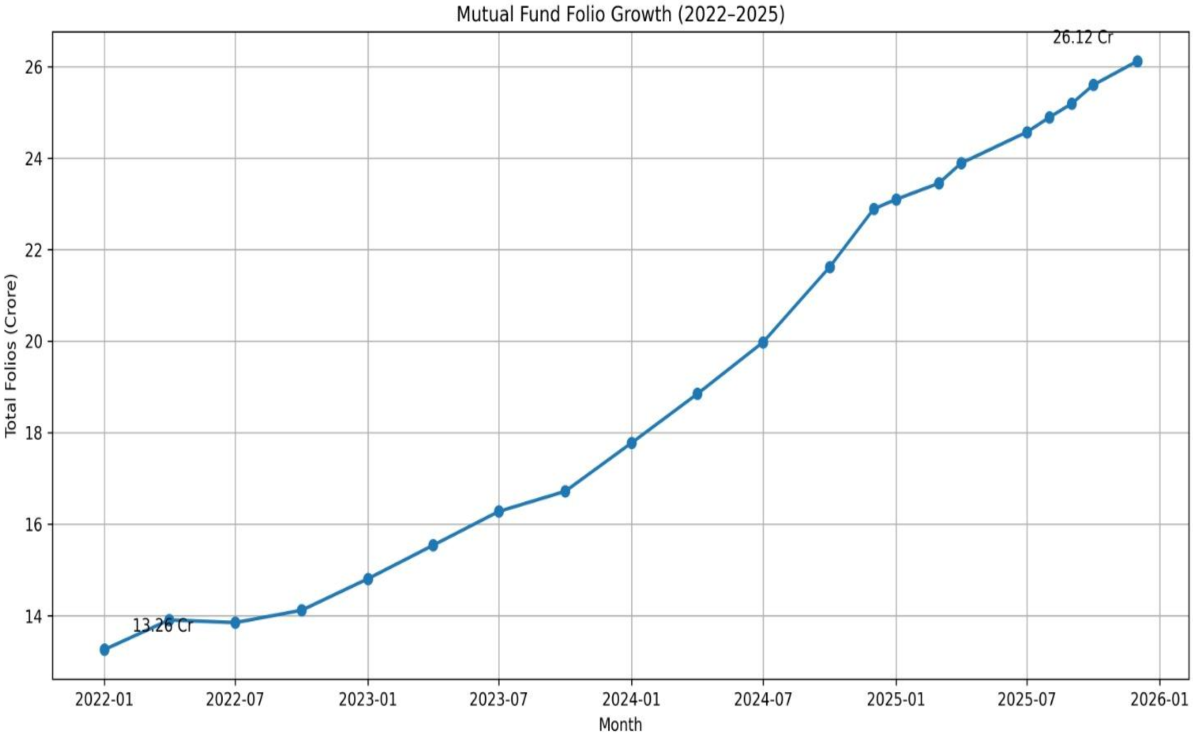


The Investor Analysis Dashboard was developed to understand investor behavior, transaction patterns, demographic characteristics, and investment preferences. The dashboard provides valuable insights into how investors interact with mutual fund products and helps identify trends in investment activity.

The dashboard analyzes investor transactions across multiple dimensions, including age group, gender, city tier, annual income, transaction type, and payment mode. Interactive visualizations allow users to explore investor participation patterns and evaluate differences across demographic segments.

Investor behavior analysis is essential for understanding customer preferences and designing investment products that align with investor needs. The dashboard helps identify key investor segments and supports data-driven customer engagement strategies.

Portfolio Analysis Dashboard



The Portfolio Analysis Dashboard provides a detailed view of mutual fund portfolio composition and sector allocation. The dashboard was designed to help investors understand how fund assets are distributed across industries, individual securities, and portfolio holdings.

Portfolio diversification is one of the most important factors influencing investment risk. The dashboard enables users to analyze sector-wise allocation, top holdings, portfolio concentration, and overall diversification levels. By visualizing portfolio composition, investors can evaluate whether a fund is adequately diversified or heavily concentrated in specific sectors or securities.

The dashboard also complements the Herfindahl-Hirschman Index (HHI) analysis by providing a visual representation of portfolio concentration. Users can quickly identify funds with high exposure to specific sectors and assess the potential impact of market fluctuations on portfolio performance.

Key Findings and Recommendations

The Bluestock Mutual Fund Analytics project revealed key insights across fund performance, risk management, investor behavior, and portfolio construction.

1. Fund Performance Insights

- Several mutual fund schemes consistently outperformed their benchmark indices across multiple investment horizons.
- Funds with higher Sharpe Ratios delivered stronger risk-adjusted returns, highlighting the importance of evaluating returns relative to risk.
- Long-term investments generally exhibited more stable and predictable performance than shorter investment periods, reinforcing the benefits of long-term wealth creation strategies.

2. Risk Analytics Insights

- Small-cap funds showed the highest Value at Risk (VaR) and Conditional Value at Risk (CVaR), indicating greater vulnerability to adverse market movements.
- Rolling Sharpe Ratio analysis demonstrated that risk-adjusted performance fluctuates significantly across different market conditions.
- Advanced risk metrics such as VaR, CVaR, and rolling performance measures provided deeper insights than return-based analysis alone.

3. Investor Behavior Insights

- The 2024 investor cohort generated the highest overall investment volume.
- Systematic Investment Plans (SIPs) emerged as the most preferred investment method among investors.
- Individuals aged 26–55 represented the largest investor segment, indicating strong participation from the working-age population.
- Tier-1 cities contributed a substantial share of total investment activity, reflecting higher investment penetration in metropolitan regions.

4. Portfolio Analysis Insights

- HHI (Herfindahl–Hirschman Index) analysis identified certain mutual funds with relatively high portfolio concentration, suggesting increased exposure to specific holdings.
- Large-cap funds generally maintained better diversification compared to small-cap funds.
- Sector allocation played a critical role in determining portfolio risk characteristics and overall diversification levels.

Conclusion and Future Enhancements

The Bluestock Mutual Fund Analytics project successfully developed a comprehensive analytics platform for evaluating mutual fund performance, risk characteristics, investor behavior, and portfolio composition. The project integrated multiple datasets and applied a structured ETL process to transform raw data into meaningful analytical insights.

Through Exploratory Data Analysis (EDA), performance evaluation, advanced risk metrics, investor cohort analysis, SIP continuity assessment, portfolio concentration measurement, and dashboard visualization, the project achieved all defined objectives. The implementation demonstrated how data analytics can support informed investment decision-making and improve financial analysis capabilities.

Advanced techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio Analysis, Herfindahl-Hirschman Index (HHI), and fund recommendation systems provided deeper insights into mutual fund performance and risk exposure. Interactive Power BI dashboards further enhanced data accessibility by presenting complex analytical results through intuitive visualizations.

Overall, the project highlights the importance of data-driven investment analysis and demonstrates the practical application of data analytics, business intelligence, and financial risk management concepts in the mutual fund industry.